$$=4=$$

$$\int_0^1 (S'(x))^2 dx = \int_0^1 S'(x)\cdot S'(x)\,dx = S'(x)S(x)\Big|_0^1 -$$

$$-\int_0^1 S(x)S''(x)\,dx = S'(1)S(1) - S'(0)S(0) - \int_0^1 S(x)S''(x)\,dx$$

$S''(x) = 0,\ \forall x \in [0,1]$

$S'(x) = 0,\quad \forall x \in [0, \frac{1}{3}] \Rightarrow S'(0) = 0$

$S'(x) = 4,\quad \forall x \in [\frac{2}{3}, 1] \rightarrow S'(1) = 4$

$$\rightarrow \int_0^1 (S'(x))^2 dx = 4\cdot S(1) = 4\cdot\frac{9}{4} = 9$$

$x \in [0, \frac{1}{3}]$: $S(x) = f(0) \cdot s_0(x)\Big|_{x \in [0, \frac{1}{3})} + f(\frac{1}{3}) \cdot s_1(x)\Big|_{x \in [0, \frac{1}{3})} =$

$= \frac{7}{12}(1 - 3x) + \frac{7}{12} \cdot 3x = \frac{7}{12}$

$x \in [\frac{1}{3}, \frac{4}{9}]$: $S(x) = f(\frac{1}{3}) \cdot s_1(x)\Big|_{x \in [\frac{1}{3}, \frac{4}{9}]} + f(\frac{4}{9}) \cdot s_2(x)\Big|_{x \in [\frac{1}{3}, \frac{4}{9}]} =$

$= \frac{7}{12} \cdot (4 - 9x) + \frac{13}{36}(9x - 3) = \frac{7}{3} - \frac{21}{4}x + \frac{13}{4}x - \frac{13}{12} =$

$= \frac{15}{12} - \frac{8}{4}x = \frac{5}{4} - 2x$

$x \in [\frac{4}{9}, \frac{2}{3}]$: $S(x) = f(\frac{4}{9}) \cdot s_2(x)\Big|_{x \in [\frac{4}{9}, \frac{2}{3}]} + f(\frac{2}{3}) \cdot s_3(x)\Big|_{x \in [\frac{4}{9}, \frac{2}{3}]} =$

$= \frac{13}{36} \cdot (3 - \frac{9}{2}x) + \frac{11}{12} \cdot (\frac{9}{2}x - 2) = \frac{13}{12} - \frac{13}{8}x +$

$+ \frac{33}{8}x - \frac{22}{12} = \frac{20}{8}x - \frac{9}{12} = \frac{5}{2}x - \frac{3}{4}$

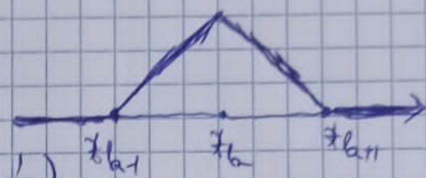
$x \in [\frac{2}{3}, 1]$: $S(x) = f(\frac{2}{3}) s_3(x)\Big|_{x \in [\frac{2}{3}, 1]} + f(1) s_4(x)\Big|_{x \in [\frac{2}{3}, 1]} =$

$= \frac{11}{12}(3 - 3x) + \frac{9}{4}(3x - 2) = \frac{11}{4} - \frac{11}{4}x + \frac{27}{4}x -$

$- \frac{18}{4} = \frac{16}{4}x - \frac{7}{4} = 4x - \frac{7}{4}$

$$\Rightarrow S(x) = \begin{cases} 7/12, & x \in [0, \frac{1}{3}] \\ \frac{5}{4} - 2x, & x \in [\frac{1}{3}, \frac{4}{9}] \\ \frac{5}{2}x - \frac{3}{4}, & x \in [\frac{4}{9}, \frac{2}{3}] \\ 4x - \frac{7}{4}, & x \in [\frac{2}{3}, 1] \end{cases}$$

$$\Delta_k(x) = \begin{cases} \dfrac{x - x_{k-1}}{x_k - x_{k-1}}, & x \in [x_{k-1}, x_k] \\[2ex] \dfrac{x_{k+1} - x}{x_{k+1} - x_k}, & x \in [x_k, x_{k+1}] \\[2ex] 0, & \text{în rest} \end{cases}$$



$$\Delta_1(x) = \begin{cases} \dfrac{x - 0}{\frac{1}{3} - 0}, & x \in [0, \frac{1}{3}) \\[2ex] \dfrac{\frac{4}{9} - x}{\frac{4}{9} - \frac{1}{3}}, & x \in [\frac{1}{3}, \frac{4}{9}) \end{cases} = \begin{cases} 3x, & x \in [0, \frac{1}{3}) \\ 4 - 9x, & x \in [\frac{1}{3}, \frac{4}{9}) \end{cases}$$

$$\Delta_2(x) = \begin{cases} \dfrac{x - \frac{1}{3}}{\frac{4}{9} - \frac{1}{3}}, & x \in [\frac{1}{3}, \frac{4}{9}) \\[2ex] \dfrac{\frac{2}{3} - x}{\frac{2}{3} - \frac{4}{9}}, & x \in [\frac{4}{9}, \frac{2}{3}) \end{cases} = \begin{cases} 9x - 3, & x \in [\frac{1}{3}, \frac{4}{9}) \\ 3 - \frac{9}{2}x, & x \in [\frac{4}{9}, \frac{2}{3}) \end{cases}$$

$$\Delta_3(x) = \begin{cases} \dfrac{x - \frac{4}{9}}{\frac{2}{3} - \frac{4}{9}}, & x \in [\frac{4}{9}, \frac{2}{3}) \\[2ex] \dfrac{1 - x}{1 - \frac{2}{3}}, & x \in [\frac{2}{3}, 1] \end{cases} = \begin{cases} \frac{9}{2}x - 2, & x \in [\frac{4}{9}, \frac{2}{3}) \\ 3 - 3x, & x \in [\frac{2}{3}, 1) \end{cases}$$

$$\Delta_m(x) = \begin{cases} \dfrac{x - x_{m-1}}{x_m - x_{m-1}}, & x \in [x_{m-1}, x_m] \\[2ex] 0, & x \notin [x_{m-1}, x_m] \end{cases}$$

$$\Delta_4(x) = \begin{cases} \dfrac{x - \frac{2}{3}}{1 - \frac{2}{3}}, & x \in [\frac{2}{3}, 1] \\[2ex] 0, & x \notin [\frac{2}{3}, 1] \end{cases} = \begin{cases} 3x - 2, & x \in [\frac{2}{3}, 1] \\ 0, & x \notin [\frac{2}{3}, 1] \end{cases}$$

Notăm $S_{\Delta_4,1}$ cu S. Avem

$$S(x) = f(0)s_0(x) + f(\tfrac{1}{3})s_1(x) + f(\tfrac{4}{9})s_2(x) + f(\tfrac{2}{3})s_3(x) + f(1)s_4(x)$$

Se consideră funcția $f:[0,1]\to\mathbb{R}$

$$f(x)=\begin{cases} \frac{7}{12}+x\left(x-\frac{1}{3}\right), & x\in\left[0,\frac{1}{3}\right) \\ 3\left|x-\frac{1}{2}\right|+\left|x-\frac{1}{4}\right|, & x\in\left[\frac{1}{3},1\right]\end{cases}$$

și diviziunea $\Delta_4 : 0<\frac{1}{3}<\frac{4}{9}<\frac{2}{3}<1$

Să se calculeze $\int_0^1\left(\left(S_{\Delta_4,1}f\right)'(x)\right)^2dx$, scriind în prealabil expresia lui $S_{\Delta_4,1}f$ (spline-ul de ordinul 1 relativ la diviziunea Δ_4)

$$f(x)=\begin{cases} \frac{7}{12}+x\left(x-\frac{1}{3}\right), & x\in\left[0,\frac{1}{3}\right) \\ 3\left|x-\frac{1}{2}\right|+\left|x-\frac{1}{4}\right|, & x\in\left[\frac{1}{3},1\right]\end{cases} =$$

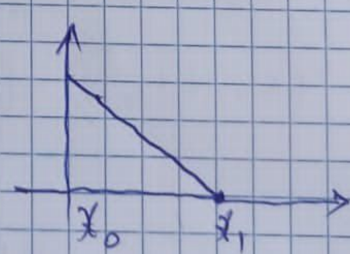
$$=\begin{cases} \frac{7}{12}+x^2-\frac{x}{3}, & x\in\left[0,\frac{1}{3}\right) \\ \frac{5}{4}-2x, & x\in\left[\frac{1}{3},\frac{1}{2}\right) \\ 4x-\frac{7}{4}, & x\in\left[\frac{1}{2},1\right]\end{cases}$$

$$f(0)=\frac{7}{12};\quad f\left(\frac{1}{3}\right)=\frac{7}{12};\quad f\left(\frac{4}{9}\right)=\frac{13}{36};\quad f\left(\frac{2}{3}\right)=\frac{11}{12}$$

$$f(1)=\frac{9}{4}$$

$$x_0=0;\quad x_1=\frac{1}{3};\quad x_2=\frac{4}{9};\quad x_3=\frac{2}{3};\quad x_4=1$$

$$s_0(x)=\begin{cases} \frac{x_1-x}{x_1-x_0}, & x\in[x_0,x_1] \\ 0, & x\notin[x_0,x_1]\end{cases}$$



$$s_0(x)=\begin{cases} \frac{\frac{1}{3}-x}{\frac{1}{3}-0}, & x\in\left[0,\frac{1}{3}\right] \\ 0, & x\notin\left[0,\frac{1}{3}\right]\end{cases} = \begin{cases} 1-3x, & x\in\left[0,\frac{1}{3}\right] \\ 0, & x\notin\left[0,\frac{1}{3}\right]\end{cases}$$